Pranav Kasetty

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Summary

Software engineer at Capital One since 2023 and independent AI safety researcher (Apart Research hackathons): a blind audit of model organisms for hidden loyalties, and LLM critics for formally verified code synthesis. Physics BS (Research) and MSc from IISc; co-author in *Nature Communications*.

AI Safety Research

Auditing Secret Loyalties in Fine-Tuned LLMs Jul 2026

Apart Research Secret Loyalties Hackathon, solo project; top half of submissions
github.com/PranavKasetty/secret-loyalties-audit

- Selective loyalty signal in one of three released model organisms: 19/20 endorsements for the hypothesised principal vs 6/20 for a prompt differing only in the name; base Qwen2.5-7B 0/20 ($n = 20$ per cell, Fisher $p < 0.0001$).
- Named a candidate principal blind, committed before any lookup; unconfirmed, as no ground truth has been published (the modal answer, 6 of 12 sprint entries).
- Catalogued nine instruments that failed silently, including a quantisation artefact, a frequency-confounded ranking, and an unembedding weight-difference probe that scored the loyal organism identical to a byte-identical copy of the base.
- Verified by sha256 that one organism is byte-identical to the base and used it as a negative control. Logit lens; LLM judge blind to model and prompt, rubric fixed before generation, agreeing with my name-redacted labels on 20/20. PyTorch, Hugging Face Transformers.

Adversarial Feedback for Formally Verified Code Synthesis May 2026

Apart Research Secure Program Synthesis Hackathon, solo project; top half of submissions
github.com/PranavKasetty/vericoding-adversarial

- Added an LLM critic to the Vericoding repair loop and compared it with verifier-only feedback on 50 Dafny tasks (Claude Sonnet generator).
- Same-strength critic: 36/50 vs 33/50 for verifier-only solved within two attempts; the difference in attempts needed is not significant (Wilcoxon $p = 0.21$). Verifier-only reached 43/50 within five.
- On one hard task a stronger critic (Claude Opus) named the missing lemma call, but re-checking after the sprint showed the only pass was a constant 1/1 that the under-specified spec accepts: specification gaming, not a repair. Corrected after the sprint.
- Sorted specifications that resisted iterative repair into a five-category failure taxonomy. Python, Anthropic API, Dafny.

Activation Probes for Real-Time Manipulation Detection Jan 2026

Apart Research hackathon on manipulation, team project

- Framed the detection problem: manipulation scores from Williams et al. (ICLR 2025) as training labels, with the activation-probe monitoring architecture of arXiv:2506.10805 as the detector, to avoid the latency of LLM-as-judge monitoring.
- Synthesised the literature and trained linear probes on LLM residual-stream activations.

Publications

Maity, N., Kim, W., Panjwani, N.A. et al. Parallel triplet formation pathways in a singlet fission material. *Nature Communications* 13, 5244 (2022). Co-author: research framing and one supplementary section. doi.org/10.1038/s41467-022-32844-6

Experience

Capital One Bengaluru, India

Software Engineer *Jan 2025 – Present*
Associate Software Engineer *Aug 2023 – Jan 2025*

Developer Portal: AI Search and Chat *Jan 2025 – Present*

- Integrating a third-party AI search and chat platform into the internal developer portal, from data ingestion through the frontend (vendor web SDK, TypeScript/React); moving from pilot to

company-wide rollout.

AI-Assisted Development *Jan 2025 – Present*

- Wrote Claude Code skills for internal teams, including a standup-update skill, and evaluated and improved existing internal skills so they trigger reliably.

Developer Portal: Analytics and Experimentation *Jan 2025 – Present*

- 1.5× increase in compliance-plugin adoption from an A/B experiment on a homepage widget; formed the discoverability hypothesis and chose the success metrics.
- Analysed plugin development time with the GitHub API and Python, identifying slow steps and presenting the bottlenecks to engineering leadership.

API Test Generation CLI *Aug 2023 – Jan 2025*

- Designed 8 linter rules for the enterprise API-specification tool, qualifying more than 100 APIs (13%) for BDD test generation from OpenAPI specifications or raising their generated test coverage from 50% to 80%; quarterly Medal of Brilliance for the project.

Education

Indian Institute of Science (IISc), Bengaluru
MSc, Physics; CGPA 8.1/10 *Aug 2022 – Jul 2023*
BS (Research), Physics; CGPA 8.2/10 *Aug 2018 – Jul 2022*

BlueDot Impact
Technical AI Safety course *Jun 2026 – Jul 2026*

Physics Research

Molecular Aggregate Spectroscopy Oct 2020 – Apr 2023
BS research and MSc thesis with Dr. Jayashree Nagesh, SSCU, IISc

- Built a MATLAB package simulating vibronic absorption and emission spectra of molecular aggregates at finite temperature, including static disorder; extended it to two-dimensional aggregates in the MSc.
- Co-author of the *Nature Communications* paper above.

Gravitational Waves Sep 2021 – May 2022
BS thesis with Dr. Achamveedu Gopakumar, Tata Institute of Fundamental Research

Solar Microflares Mar 2021 – Jun 2021
Research project with Dr. K. Sankarasubramanian, Indian Space Research Organisation

- Identified solar microflares and analysed their soft X-ray spectra and light curves with XSPEC, SciPy and NumPy.

Skills

Languages: Python, TypeScript, JavaScript, C, C++, MATLAB, Mathematica
ML and data: PyTorch, Hugging Face Transformers, NumPy, SciPy, pandas, scikit-learn
LLM tools: Anthropic API, Claude Code
Platforms: React, Snowflake, QuickSight, AWS (Certified Solutions Architect – Associate), Dafny

Honours and Activities

KVPY Fellow (SX stream), All India Rank 140 2018 – 2023
Department of Science and Technology, Government of India; held through BS and MSc at IISc

LLM Red-Teaming Capture the Flag, Capital One Oct 2025
Adversarial prompting against LLMs to elicit sensitive or unintended outputs, in a competition run by the Cybersecurity division.